University of Florida, Department of Mathematics

Topology PhD Exam, May 2000

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that there is no retraction of the Möbius band onto its boundary.

2. State and prove the Contraction Mapping Theorem.

3. Give the Mayer–Vietoris sequence for homology. Use this to compute the homology groups for the n-sphere S^n.

4. Suppose that $f, g : X \to S^n$ are two maps such that $f(x)$ and $g(x)$ are not antipodal for all $x \in X$. Show that f and g are homotopic.

5. Show that no covering space of the 2-torus has the homotopy type of a figure 8.

Answer the following with complete definitions or statements or short proofs.

6. Give a classification of the compact surfaces.

7. State Urysohn's Lemma. State the Tietze Extension Theorem.

8. State the Five Lemma.

9. Define the mapping cylinder of $f : X \to Y$. Define the mapping torus of $f : X \to X$.

10. State the exact homology sequence for a pair of spaces.

11. State the Jordan Curve Theorem.

12. Define retraction and deformation retraction.

13. State the Ascoli (Arzelà–Ascoli) Theorem.

14. Let V be a complete metric space such that no point is isolated. Can V be a countably infinite set?

15. Let A be a countable subset of $\mathbb{R}^2$. Show that $\mathbb{R}^2 \setminus A$ is arcwise connected.